

The Hubverse: Streamlining Collaborative Infectious Disease Modeling

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Consortium of Infectious Disease Modeling Hubs

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Background

✗ Where we started

Infectious disease modeling scaled rapidly through COVID-19...

- ...but early on the landscape was **fragmented**:
 - Each group used its **own data structures**
 - Results were **hard to integrate or compare**
 - **Modelers and decision-makers** not always aligned

“Comparing the accuracy of forecasting applications is difficult because forecasting methods, forecast outcomes, and reported validation metrics varied widely.”

- Chretien et al., PLOS ONE, 2014

✨ The promise of modeling hubs

Modeling hubs coordinate collaborative forecasting:

- Provide **centralised location** for effort coordination
- Define **data standards** and **modeling targets**
- Improve **transparency** and **comparability**
- Aggregate forecasts **enabling ensembles**
- Facilitate timely **public health decision-making**

***“Collaborative Hubs: Making the Most of Predictive Epidemic Modeling”**, American Journal of Public Health Reich, et al. 2022*

Project origins

- **Pre-COVID:** Forecasting code base existed for CDC **influenza** hubs
- **During COVID:** That code was reused for new **COVID-19** hubs + demand internationally (e.g. Europe) for similar setups
- **!** Problem: Each hub required **manual editing** of source code

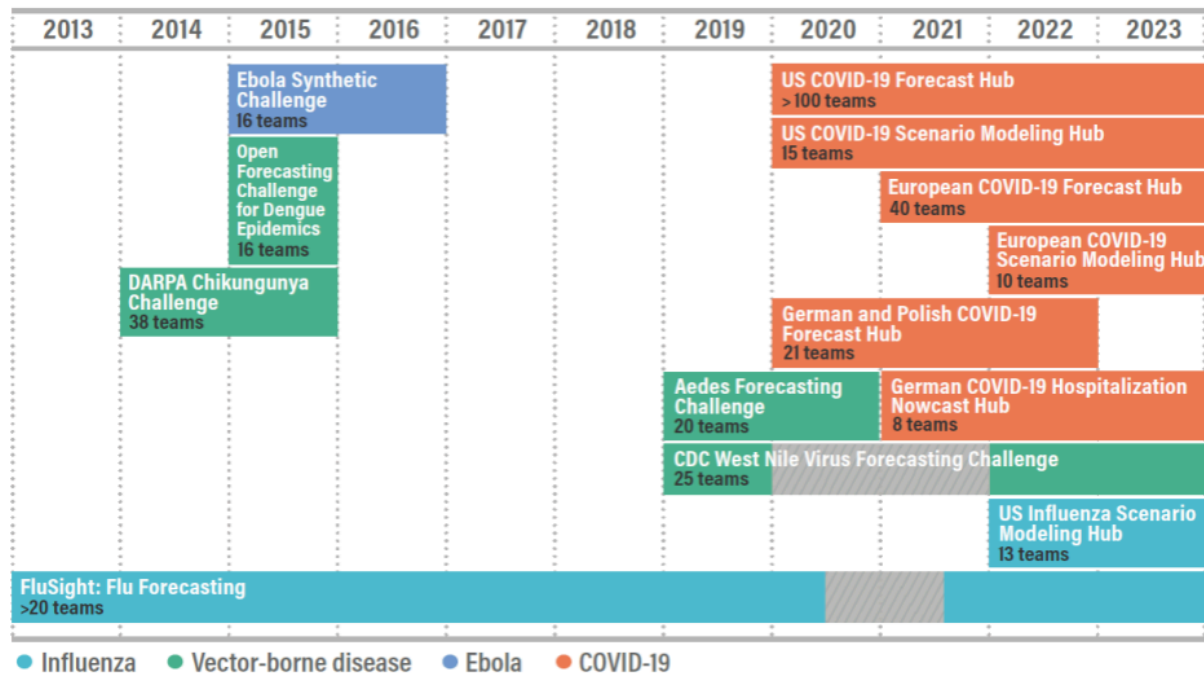


Figure credits: Alex Vespignani and Nicole Samay

Enter the hubverse

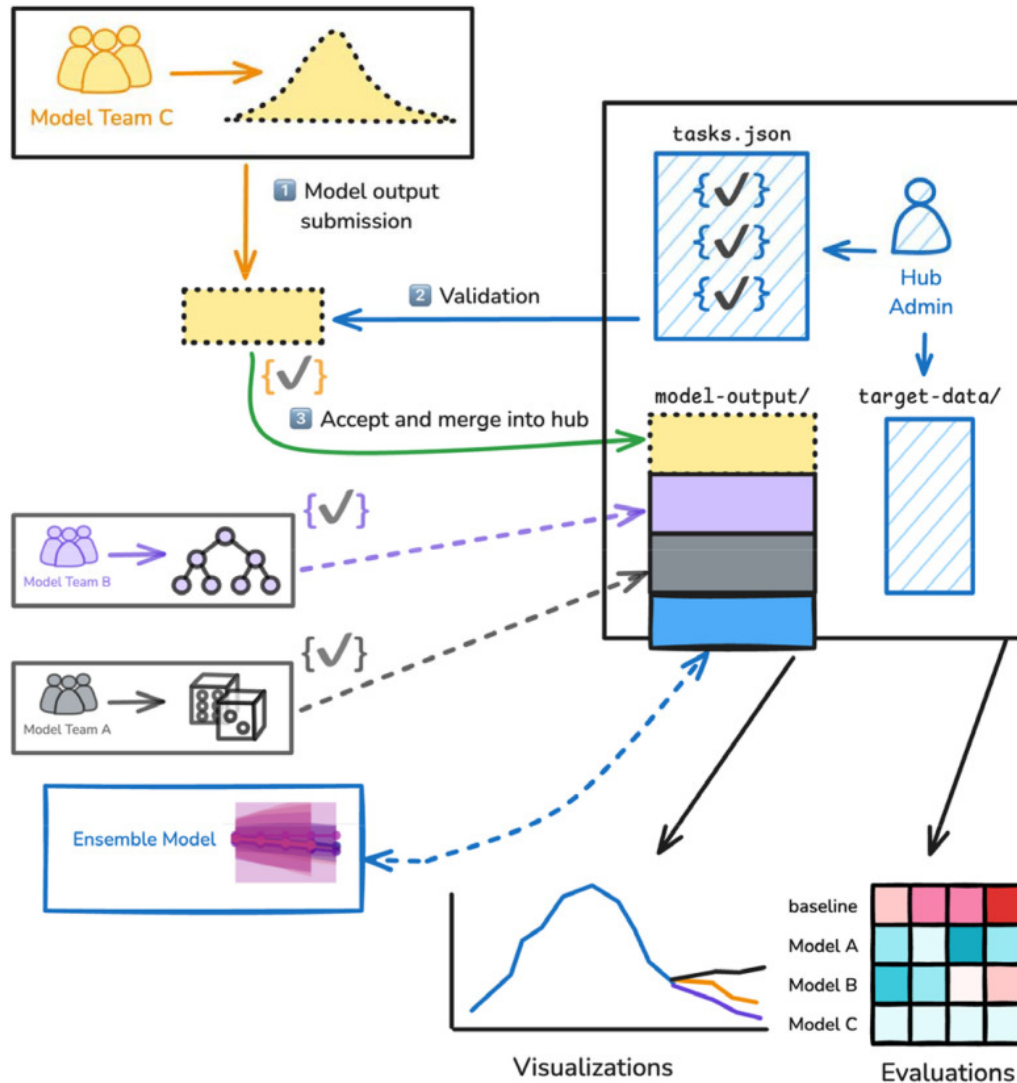
An open-source **software ecosystem** to power modeling hubs:

- GitHub repositories for **centralising hub activity**
- **Data standards** for infectious disease modeling data
- **Schema-driven configuration** for modeling tasks + hub setup
- Modular tools for validation, access, evaluation, ensembling, communication and hub administration
- **Supports full lifecycle:** from hub set up, data submission to decision-making

 A software platform for collaborative infectious disease modelling (2026), *Nature Health, Consortium of Infectious Disease Modeling Hubs et al.*
<https://doi.org/10.1038/s44360-026-00145-7>

Anatomy of a hub

What a hub actually is



1. Modeling teams **submit model output**
2. Submissions **validated** against `tasks.json`
3. Accepted output **merged into the hub**

Everything downstream, **ensembles, visualisations, evaluations**, comes for free once the data are standardised.

Figure 1 from the [hubverse paper](#) (preprint, CC BY 4.0)

A shared data standard

Modeling hubs are **built around a shared data standard**:

- **Structured hub layout**: consistent file system for organizing hub materials
- **Standard model output format**: for file content and naming
- **Modeling task definition**: what's predicted, by when, for where, in what form

 Enables comparability, validation, and streamlined data access

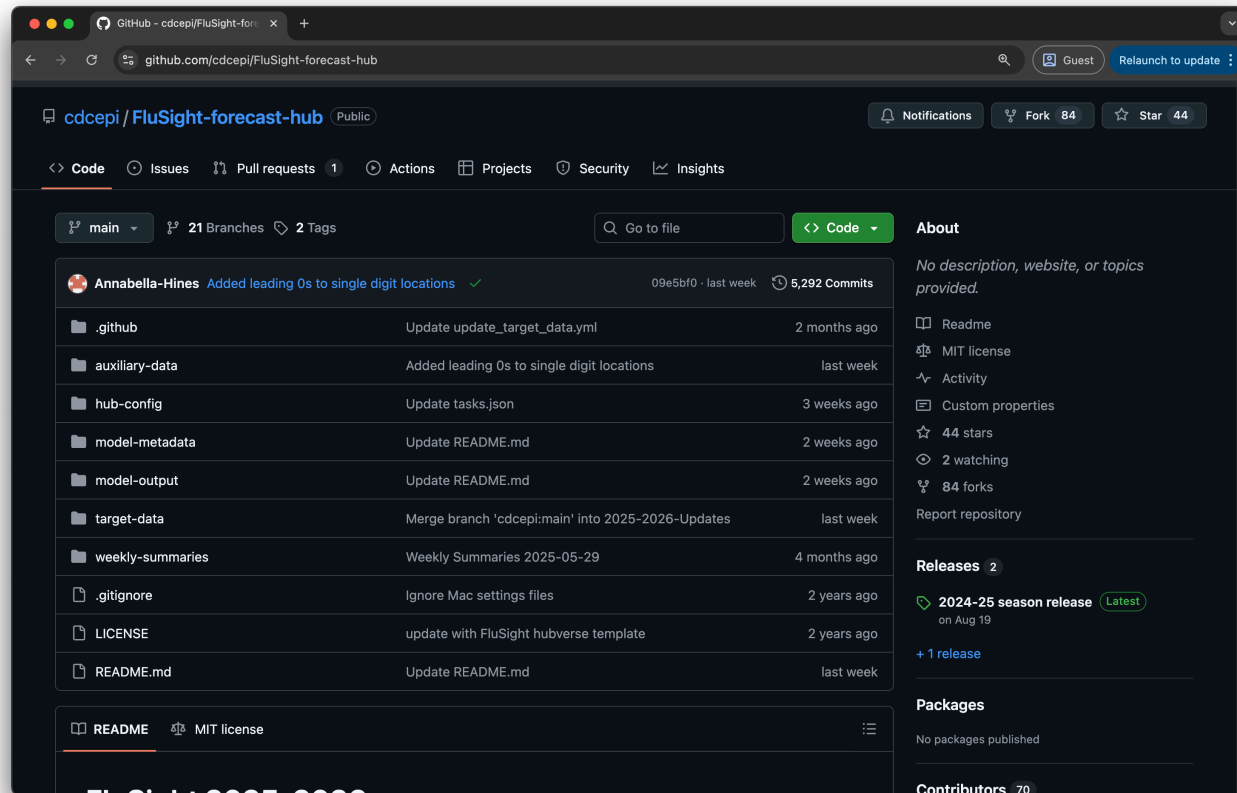
```
1 FluSight-forecast-hub/
2 |— hub-config/
3 |   |— admin.json
4 |   |— tasks.json
5 |   |— model-metadata-schema.json
6 |   |— target-data.json
7 |— model-output/
8 |   |— <model_id>/
9 |       |— <round_id>-<model_id>.csv
10 |— model-metadata/
11 |   |— <model_id>.yaml
12 |— target-data/
```



Meet the hub: CDC FluSight

The hub we'll follow: CDC FluSight

 <https://github.com/cdcepi/FluSight-forecast-hub>



- Used by US CDC to **monitor influenza severity**
- In season, **~60 models** from **~30 teams**, every single week
- **95 models** from **47 teams** across **88 rounds** since 2023
- Hosted on **GitHub + S3 cloud mirror**
- Managed with the **full hubverse stack** since **2023/24**



Where model output lives

`model-output/<model_id>/<round_id>—<model_id>.csv` · one directory per model, one file per round

The screenshot shows a GitHub repository for `FluSight-forecast-hub` by `cdcepi`. The left sidebar shows the file tree with the `model-output` directory selected. A note indicates "One folder per model". The main content area shows the `model-output/CFA_Pyrenew-Pyrenew_E_Flu` directory. A table lists the files in this directory, showing the round ID, model ID, and the last commit message and date.

Round ID	Model ID	Last commit message	Last commit date
2025-11-22	CFA_Pyrenew-Pyrenew_E_Flu.csv	add e forecasts	10 months ago
2025-11-29	CFA_Pyrenew-Pyrenew_E_Flu.csv	Update PyRenew forecasts table for 2025-11-26	10 months ago
2025-12-06	CFA_Pyrenew-Pyrenew_E_Flu.csv	Update PyRenew forecasts table for 2025-12-03	10 months ago
2025-12-13	CFA_Pyrenew-Pyrenew_E_Flu.csv	Revised Pyrenew submission	9 months ago
2025-12-20	CFA_Pyrenew-Pyrenew_E_Flu.csv	Update PyRenew forecasts table for 2025-12-17	9 months ago
2025-12-27	CFA_Pyrenew-Pyrenew_E_Flu.csv	Update PyRenew forecasts table for 2025-12-24	9 months ago
2026-01-03	CFA_Pyrenew-Pyrenew_E_Flu.csv	Update PyRenew forecasts table for 2026-01-02	9 months ago
2026-01-10	CFA_Pyrenew-Pyrenew_E_Flu.csv	Update PyRenew forecasts table for 2026-01-07	8 months ago
2026-01-17	CFA_Pyrenew-Pyrenew_E_Flu.csv	Update PyRenew forecasts table for 2026-01-14	8 months ago
2026-01-24	CFA_Pyrenew-Pyrenew_E_Flu.csv	Update PyRenew forecasts table for 2026-01-21	8 months ago
2026-01-31	CFA_Pyrenew-Pyrenew_E_Flu.csv	Update PyRenew forecasts table for 2026-01-28	8 months ago

Configuring FluSight

Config-driven hub setup

Hub administrators configure hubs with structured JSON config files:

- `admin.json`: hub-level metadata
- `tasks.json`: the **scientific question**: what to predict, when, and how
- `model-metadata-schema.json`: what modelers must tell you about their model
- `target-data.json`: what **observed data** should look like

All validated against a **versioned JSON schema**. The config *is* the contract between administrators, modelers and tooling.



admin.json: the hub's identity card

```
1 {
2   "schema_version": ".../schemas/main/v6.0.0/admin-schema.json",
3   "name": "US CDC FluSight",
4   "maintainer": "US CDC",
5   "contact": {
6     "name": "Rebecca Borchering",
7     "email": "xhq2@cdc.gov"
8   },
9   "repository": {
10    "host": "github",
11    "owner": "cdcepi",
12    "name": "FluSight-forecast-hub"
13  },
14  "file_format": ["csv", "parquet"],
15  "timezone": "US/Eastern",
16  "cloud": {
17    "enabled": true,
18    "host": {
19      "name": "aws",
20      "storage_service": "s3",
21      "storage_location": "cdcepi-flusight-forecast-hub"
22    }
23  }
24 }
```

Inside tasks.json

```
1 tasks.json
2 └─ rounds[]                                # a submission cycle
3     └─ round_id_from_variable: true        # round IDs live in the data...
4     └─ round_id: "reference_date"         # ...in this task ID
5     └─ submissions_due                    # window, relative to the round
6     └─ model_tasks[]                     # groups of related predictions
7         └─ task_ids{}                     # the DIMENSIONS of a prediction
8         └─ output_type{}                  # the FORM the prediction takes
9         └─ target_metadata[]              # what the target means
```

A hub can have **many rounds**, each with **many model tasks**, so a single hub can ask for **several different kinds of prediction** at once.

Task IDs: the dimensions of a prediction

```

1  "task_ids": {
2    "reference_date": {
3      "required": null,
4      "optional": ["2023-10-07", "2023-10-14", "..."]
5    },
6    "target": {
7      "required": null,
8      "optional": ["wk inc flu hosp"]
9    },
10   "horizon": {
11     "required": null,
12     "optional": [-1, 0, 1, 2, 3]
13   },
14   "location": {
15     "required": null,
16     "optional": ["US", "01", "02", "..."]
17   },
18   "target_end_date": {
19     "required": null,
20     "optional": ["2023-09-23", "2023-09-30", "..."]
21   }
22 }
```

$\text{reference_date} + \text{horizon} \times 1 \text{ week} = \text{target_end_date}$

Targets carry more than a name

`target` is a special task ID: every value in the column gets its own metadata entry

```
1 "target_metadata": [{  
2   "target_id": "wk inc flu hosp",  
3   "target_name": "incident influenza  
4     hospitalizations",  
5   "description": "Count of new hospital  
6     admissions in the week ending...",  
7   "target_units": "count",  
8   "is_step_ahead": true,  
9   "time_unit": "week",  
10  "target_type": "continuous"  
11  }]
```

Descriptive

- `target_name`, `description`: for humans, and for axis labels

Quantitative

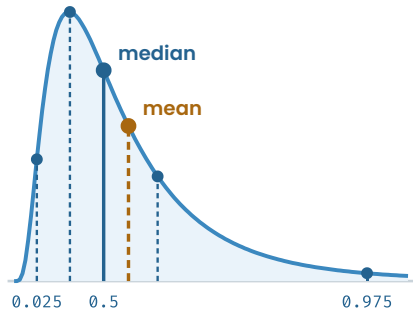
- `target_units`: what is being counted
- `is_step_ahead`, `time_unit`: one step in a sequence
- `target_type`: its statistical type (`continuous`, `ordinal`, `date`, ...)

`target_type` constrains **how a prediction of this target can be expressed**. Which is exactly what an **output type** is...

Output types: how a prediction is expressed

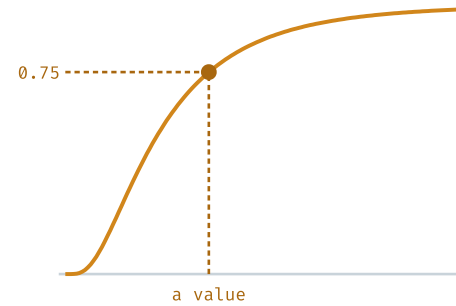
quantile · mean · median

the distribution, and two point estimates



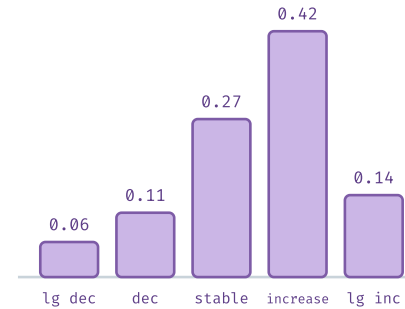
cdf

probability at or below a value



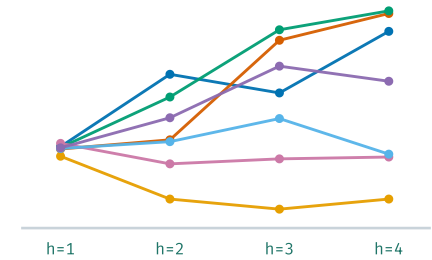
pmf

probability of each category



sample

draws from a joint distribution



output_type	output_type_id	value
mean	(none)	mean of the predictive distribution
median	(none)	median of the predictive distribution
quantile	a probability level, e.g. 0.75	the value at that quantile
cdf	a possible value, e.g. 500	$P(\text{outcome} \leq 500)$
pmf	a possible category, e.g. "increase"	$P(\text{outcome} = \text{"increase"})$
sample	a sample index, e.g. "s3"	one draw from the distribution



...and how a hub configures them

```
1  "output_type": {
2    "quantile": {
3      "output_type_id": {
4        "required": [0.01, 0.025, "...", 0.975, 0.99]
5      },
6      "is_required": true,
7      "value": { "type": "double", "minimum": 0 }
8    },
9    "sample": {
10     "output_type_id_params": {
11       "compound_taskid_set": ["reference_date", "location", "target"],
12       "min_samples_per_task": 100,
13       "max_samples_per_task": 100
14     },
15     "is_required": false,
16     "value": { "type": "integer", "minimum": 0 }
17   }
18 }
```

Configured model output in the wild

Real FluSight rows from round 2026-01-10, national. task IDs · output type · value

reference_date	horizon	target	target_end_date	location	output_type	output_type_id	value
2026-01-10	1	wk inc flu hosp	2026-01-17	US	quantile	0.01	18429
2026-01-10	1	wk inc flu hosp	2026-01-17	US	quantile	0.025	20177
2026-01-10	1	wk inc flu hosp	2026-01-17	US	quantile	0.05	21981
...							
2026-01-10	1	wk inc flu hosp	2026-01-17	US	quantile	0.5	38936
...							
2026-01-10	1	wk inc flu hosp	2026-01-17	US	quantile	0.975	61292
2026-01-10	1	wk inc flu hosp	2026-01-17	US	quantile	0.99	68125
2026-01-10	0	wk inc flu hosp	2026-01-10	US	sample	us_s1	36268
2026-01-10	1	wk inc flu hosp	2026-01-17	US	sample	us_s1	36397
2026-01-10	2	wk inc flu hosp	2026-01-24	US	sample	us_s1	33669
2026-01-10	3	wk inc flu hosp	2026-01-31	US	sample	us_s1	33727
2026-01-10	1	wk flu hosp rate change	2026-01-17	US	pmf	large_decrease	0.114
2026-01-10	1	wk flu hosp rate change	2026-01-17	US	pmf	decrease	0.244
2026-01-10	1	wk flu hosp rate change	2026-01-17	US	pmf	stable	0.158
2026-01-10	1	wk flu hosp rate change	2026-01-17	US	pmf	increase	0.231
2026-01-10	1	wk flu hosp rate change	2026-01-17	US	pmf	large_increase	0.253

Hubs collect metadata about the models too

`model-metadata-schema.json` sets out what each team must document about their model

- One file per model in `model-metadata/`, validated on every PR
- The hubverse ships a schema template, hubs adapt it

```
1 team_name: "UMass-Amherst"
2 model_abbr: "flusion"
3 model_version: "1.1"
4 model_contributors:
5   - name: "Evan Ray"
6     affiliation: "UMass Amherst"
7 license: "CC-BY-4.0"
8 data_inputs: "NHSN, FluSurv-NET and ILINet."
9 methods: "Ensemble of statistical and machine
10   learning time series models."
11 designated_model: true
```

Provenance for every prediction, and a natural basis for **model cards**.

Beyond the data standard

Automating everything we can

GitHub Actions do the operational work:

-  PR-level **model output validation**
-  **Hub configuration** validation
-  **Cloud** hub data syncing
-  **Dashboard builds**

All actions live in [hubverse-actions](#) and install with `hubCI :: use_hub_github_action()`

Cloud storage and access

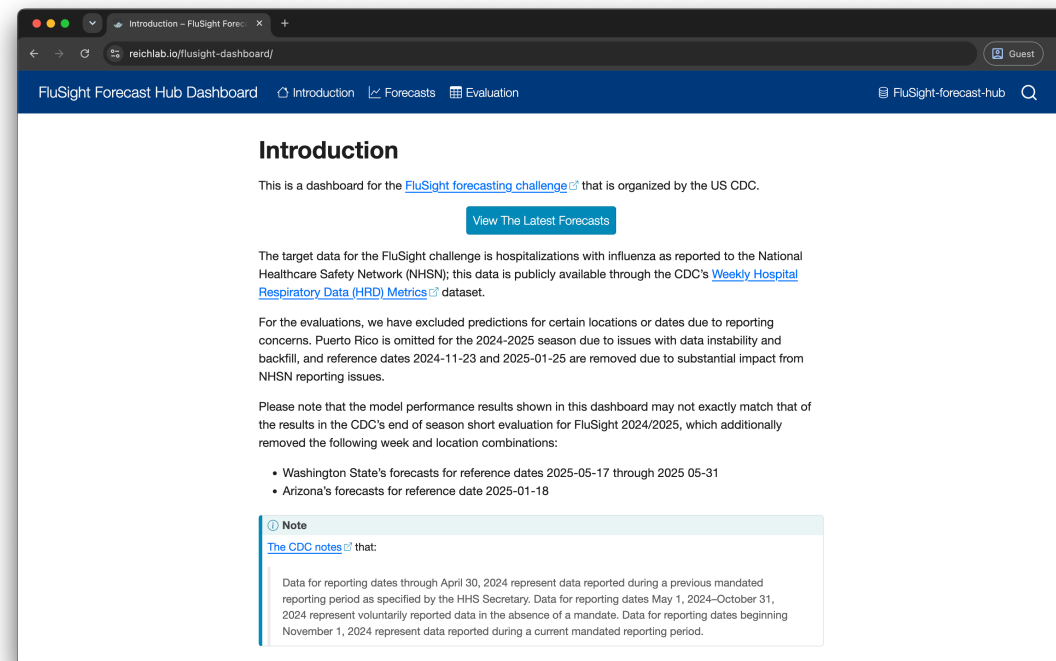
Validated hub data is mirrored to a public S3 bucket

- Opened directly as an **Arrow dataset**: no cloning, no full download
- **Query-able** via  `hubData` (R) and  `hubdata` (Python)
- Turns a hub into an **open data resource**, not just a repository

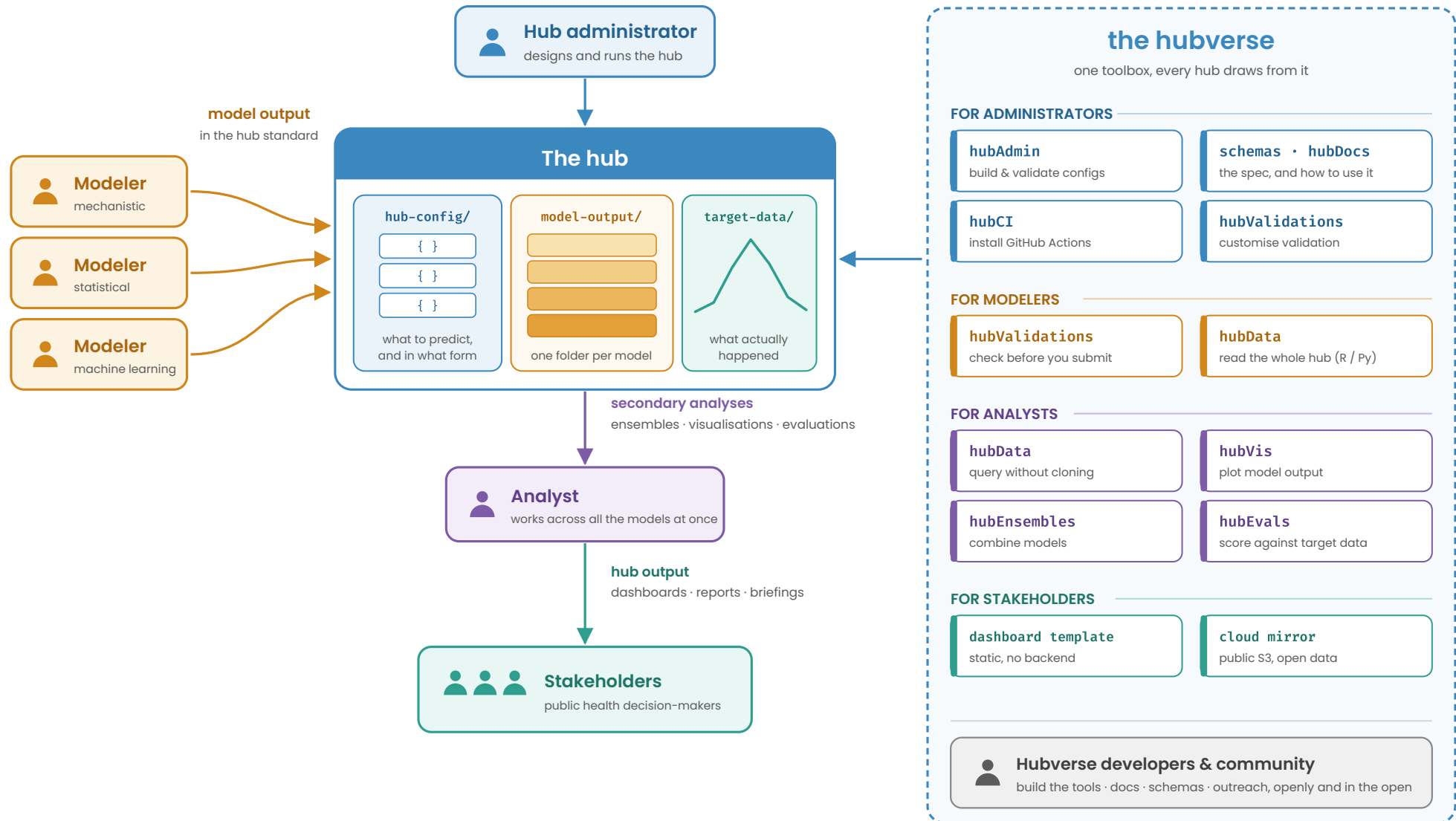
Enable it with a few lines in `admin.json`. The hubverse provisions the bucket.

Dashboards & communication

- Built with **Quarto** so easily customisable via Quarto configuration
- Deployed as a **fully static site**, no backend required
- Powered by JSON data prepared via GitHub workflows
- Interactive UI built with client-side JavaScript (fast!)
- New instances set up by copying/configuring the **hub-dashboard-template**



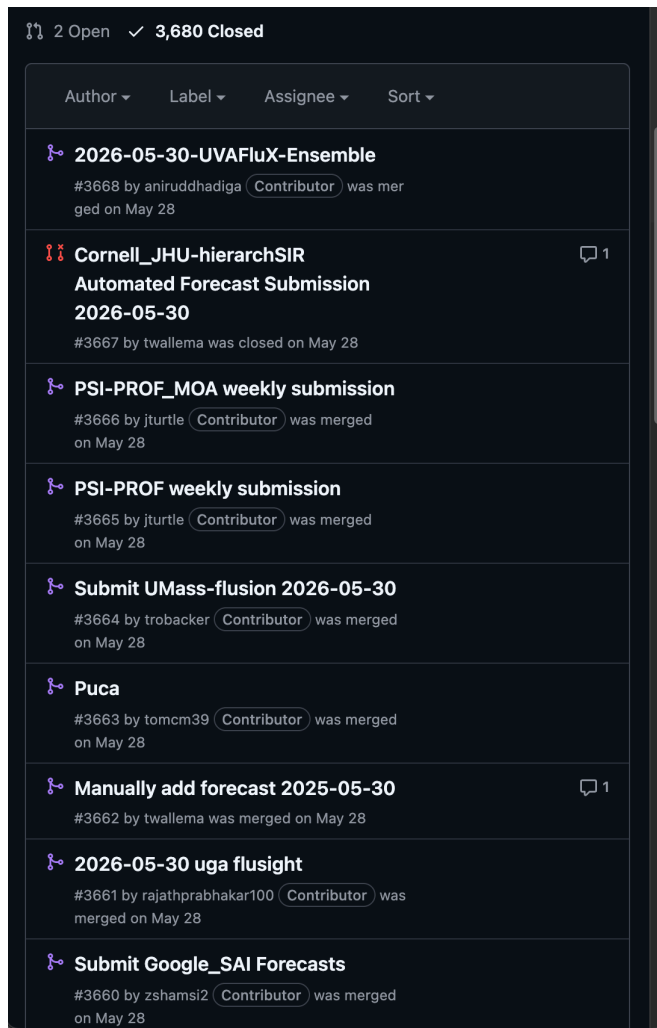
Hubverse roles and their tooling



FluSight in operation

✓ Model output validation with hubValidations

Submitted by pull request, validated by GitHub Actions, **reported back on the PR**



2 Open ✓ 3,680 Closed

Author ▾ Label ▾ Assignee ▾ Sort ▾

- 2026-05-30-UVAFluX-Ensemble**
#3668 by aniruddhadiga (Contributor) was merged on May 28
- Cornell_JHU-hierarchSIR**
Automated Forecast Submission 2026-05-30
#3667 by twallema was closed on May 28
- PSI-PROF_MOA weekly submission**
#3666 by jturtle (Contributor) was merged on May 28
- PSI-PROF weekly submission**
#3665 by jturtle (Contributor) was merged on May 28
- Submit UMass-flusion 2026-05-30**
#3664 by trobacker (Contributor) was merged on May 28
- Puca**
#3663 by tomcm39 (Contributor) was merged on May 28
- Manually add forecast 2025-05-30**
#3662 by twallema was merged on May 28
- 2026-05-30 uga flusight**
#3661 by rajathprabhakar100 (Contributor) was merged on May 28
- Submit Google_SAI Forecasts**
#3660 by zshamsi2 (Contributor) was merged on May 28

Passed

Failed

github-actions (bot) commented

Submission validation

✓ All validation checks passed.

▶ [Check results](#)

Results for commit [a1b2c3d](#)

What actually gets checked

Standard checks, generated from `tasks.json`

-  **File:** name, location, format, submission window
-  **Structure:** columns, types, duplicates, round ID
-  **Content:** valid task ID combinations, required tasks present, values in bounds
-  **Output-type specific:** quantiles don't cross, `pmf` sums to 1, sample counts
-  **Model metadata** and  **target data:** each with their own checks

Hub-specific checks in `hub-config/validations.yml`

```

1 default:
2   validate_model_data:
3     horizon_timediff:
4       fn: "opt_check_tbl_horizon_timediff"
5       pkg: "hubValidations"
6       args:
7         t0_colname: "reference_date"
8         t1_colname: "target_end_date"

```

- **Opt-in** checks shipped with `hubValidations`
- Or the hub's **own R functions**, in `src/validations/R`

`create_custom_check()` scaffolds a custom check: right structure, return classes and conventions, ready to fill in.

Accessing model output via hubData

Connect to Arrow dataset of forecast submissions

```
1 library(hubData)
2
3 hub_path <- s3_bucket(
4   "cdcepi-flusight-forecast-hub"
5 )
6 hub_con <- connect_hub(hub_path)
7 hub_con
```

```
hub_connection
9 columns
reference_date: date32[day]
target: string
horizon: int32
target_end_date: date32[day]
location: string
output_type: string
output_type_id: string
value: double
model_id: string
```

Query and collect data

```
1 # Filter for one model and forecast date using dplyr
2 library(dplyr)
3 hub_con |>
4   filter(
5     model_id == "CADPH-FluCAT_Ensemble",
6     target_end_date == "2023-10-28"
7   ) |>
8   collect_hub()
```

```
# A tibble: 92 × 9
  model_id    reference_date target horizon target_end_date location output_type
* <chr>      <date>      <chr>   <int> <date>      <chr>    <chr>
1 CADPH-FluC... 2023-10-21    wk in...     1 2023-10-28    06    quantile
2 CADPH-FluC... 2023-10-21    wk in...     1 2023-10-28    06    quantile
3 CADPH-FluC... 2023-10-21    wk in...     1 2023-10-28    06    quantile
4 CADPH-FluC... 2023-10-21    wk in...     1 2023-10-28    06    quantile
5 CADPH-FluC... 2023-10-21    wk in...     1 2023-10-28    06    quantile
# i 87 more rows
# i 2 more variables: output_type_id <chr>, value <dbl>
```

See more in [Accessing data vignette](#).

Python analogue **hub-data** also available.

Ensembling with hubEnsembles

Combine models using simple or weighted rules

```

1 forecast_df <- hub_con |>
2   filter(
3     model_id %in%
4       c(
5         "CADPH-FluCAT_Ensemble",
6         "CEPH-Rtrend_fluH",
7         "CFA_Pyrenew-Pyrenew_HE_Flu"
8       ),
9     output_type == "quantile"
10  ) |>
11  collect_hub()
12
13
14 hubEnsembles::simple_ensemble(
15   forecast_df,
16   agg_fun = median,
17   model_id = "simple-ensemble-median"
18 )

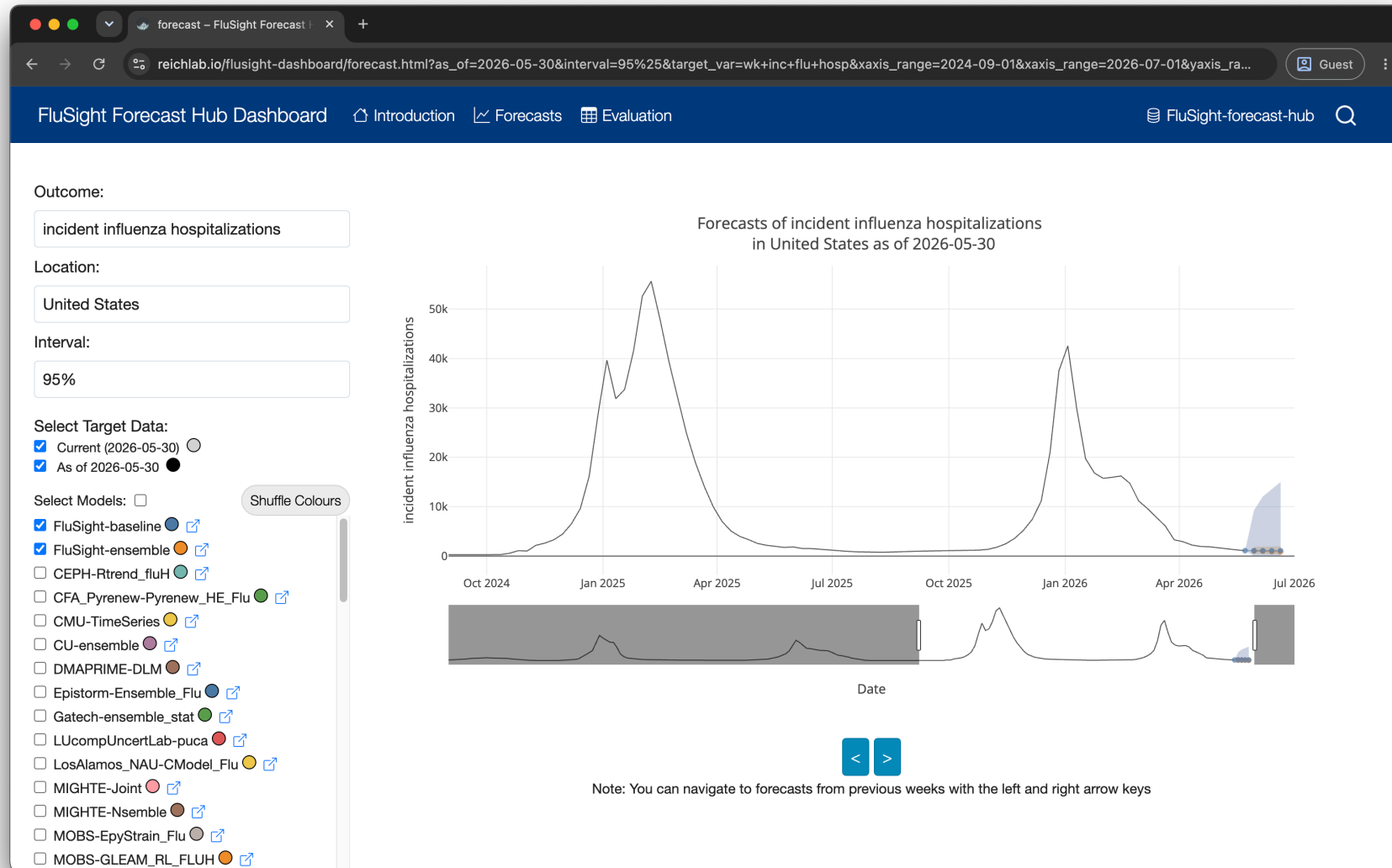
```

```

# A tibble: 492,476 × 9
  model_id    reference_date target horizon target_end_date location output_type
* <chr>      <date>          <chr>   <int> <date>          <chr>   <chr>
1 simple-ens... 2023-10-14      wk in...   -1 2023-10-07      01    quantile
2 simple-ens... 2023-10-14      wk in...   -1 2023-10-07      01    quantile
3 simple-ens... 2023-10-14      wk in...   -1 2023-10-07      01    quantile
4 simple-ens... 2023-10-14      wk in...   -1 2023-10-07      01    quantile
5 simple-ens... 2023-10-14      wk in...   -1 2023-10-07      01    quantile
# i 492,471 more rows
# i 2 more variables: output_type_id <chr>, value <dbl>

```


Dashboard – forecasts



Dashboard – model evaluations

Evaluates forecasts against target (observed) data.

Table

Plot

FluSight Forecast Hub Dashboard

Introduction Forecasts Evaluation

FluSight-forecast-hub

General Options

Target: incident influenza hospitalizations

Evaluation set: 2025-2026 season, state level

Transformation Definition

log
Natural logarithm after adding an offset to the values (offset = 1).

Metric Definitions

Rel. WIS
Relative skill of the weighted interval score, scaled in comparison to the baseline model. Compares Model M's average WIS to all other models', for all possible forecasts, then divides that by the same score for the baseline for easy comparison. Smaller values are better, and the baseline always has a value of 1.0.

WIS
Weighted interval score. The averaged interval scores (IS) of every prediction interval, plus the difference between the observed value and median. The interval score is the sum of three components: overprediction, underprediction,

Table

Model	Rel. WIS	Rel. WIS (log)	WIS	WIS (log)	Rel. MAE	Rel. MAE (log)	MAE	MAE (log)	50% Cov.	95% Cov.
Epistorm-Ensemble_Flu	0.57	0.59	12.9	0.3	0.63	0.68	19.5	0.4	54.7	95.1
CMU-TimeSeries	0.59	0.58	59.7	0.3	0.70	0.70	92.2	0.5	48.4	95.1
Google_SAI-FluEns	0.61	0.56	60.6	0.3	0.72	0.67	93.1	0.5	51.5	95.1
JHU_CSSE-CSSE_Ensemble	0.61	0.61	182.9	0.3	0.73	0.74	290.2	0.5	17.0	7.0
UMass-flusion	0.62	0.59	59.9	0.3	0.70	0.68	87.3	0.5	34.9	95.1
FluSight-HJudge_ensemble	0.63	0.59	62.3	0.3	0.74	0.70	94.8	0.5	49.0	95.1
NAU-vulPES	0.63	0.61	61.3	0.3	0.75	0.74	95.9	0.5	44.9	95.1
UGA_flucast-INFLAenza	0.63	0.59	53.1	0.3	0.72	0.69	78.4	0.5	49.5	95.1
Cornell_JHU-hierarchSIR	0.64	0.82	61.5	0.4	0.73	1.01	91.2	0.7	38.4	95.1
FluSight-trained_mean	0.64	0.59	63.1	0.3	0.74	0.71	94.9	0.5	46.2	95.1
FluSight-trained_med	0.64	0.61	63.1	0.3	0.75	0.72	96.6	0.5	50.1	95.1
NAU-FourCAT	0.64	0.73	60.5	0.4	0.77	0.88	94.6	0.6	43.8	95.1
FluSight-lap_norm	0.65	0.63	63.8	0.3	0.77	0.72	99.5	0.5	58.1	95.1
UGA_flucast-Residualnet	0.65	0.64	67.6	0.3	0.78	0.78	105.4	0.5	45.1	95.1

Closing remarks

Who's using it

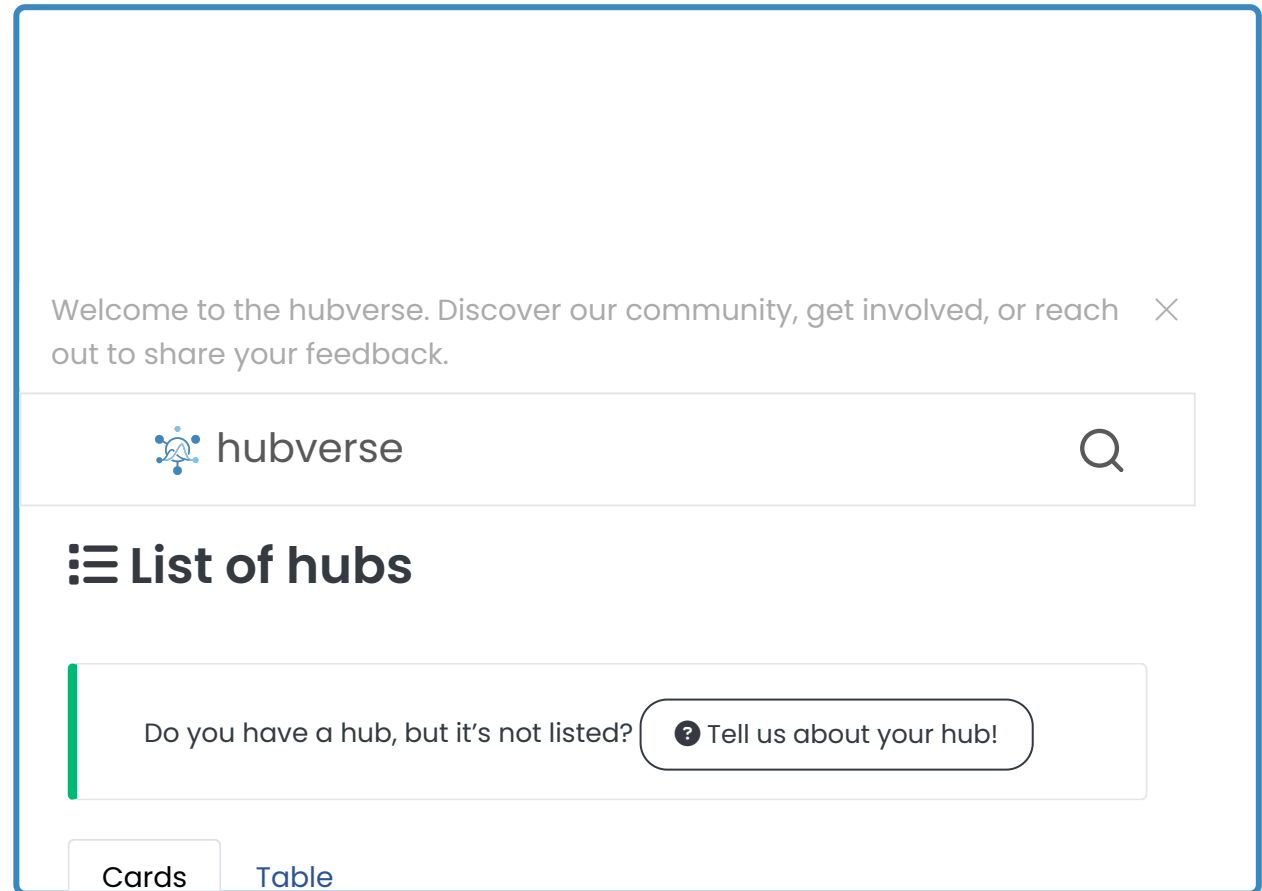
34 hubs across **15 organizations**,
on five continents

736 models · **4.6 billion rows** of
model output (*Sep 2026*)

Five kinds of hub:

-  **Community**: open, public repos
-  **Collaborative, private**:
e.g. Paraguay, Australia–Aotearoa
-  **Local / model development**: a hub on your own laptop
-  **Training**: teaching hubs
-  **Archival**: retired hubs, kept queryable

 hubverse.io/community/hubs.html



Lessons & wider relevance

-  **A good standard is leverage:** define the interface once, reuse everything downstream
-  **Standards + automation reduce friction**
-  **Configuration over code:** versioned schemas let the standard evolve without breaking hubs
-  **Open source keeps it free & accessible**
-  **Standardised, open data fuels downstream use:** teaching, reproducible research, and teams' own model development

Built on hubverse data

SISMID

Short course teaching real-world outbreak forecast evaluation, worked directly on hub data

RespiLens

A responsive web app to visualise US respiratory disease forecasts, built for state health departments and the public. ACCIDDA, UNC Chapel Hill

MicroHub workshop

Hands-on workshop: run forecast models locally in a Docker app, then stand up your own hubverse hub on GitHub. Northern Arizona University

🙏 Thank you!

- 🔗 <https://hubverse.io>
- 📖 <https://docs.hubverse.io>
- 🐙 [hubverse-org](https://github.com/hubverse-org)
- 📦 **Install the packages:** <https://hubverse-org.r-universe.dev>
- ✉️ info@r-se.eu



Members of hubverse team at the 2024 work retreat, UMass Amherst



Tip

Interested in getting involved? Check out our [Getting Involved](#) page!



Read more

Consortium of Infectious Disease Modeling Hubs *et al.* (2026)

A software platform for collaborative infectious disease modelling

Nature Health

<https://doi.org/10.1038/s44360-026-00145-7>

Open-access preprint:

<https://doi.org/10.1101/2025.10.03.25337284>